

CELLULAR SHEAF HARMONIC LATENT ALIGNMENT: TEST-TIME PROMPT ROBUSTNESS VIA RESIDUAL- STREAM TOPOLOGICAL DIFFUSION

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ABSTRACT

Large Language Models (LLMs) are extremely sensitive to semantically-neutral prompt variations—structural format edits, typos, and markup wrappers—which cause severe degradation on downstream reasoning tasks. Existing post-hoc debiasing interventions operate on late-stage output logits or apply static Euclidean vector offsets, and consequently suffer from three failure modes: mandatory clean-performance degradation ($p_{\text{clean}} < 0$), an inability to model non-linear local token interactions in intermediate representations, and high test-time latency. We introduce **Cellular Sheaf Harmonic Latent Alignment (CS-HLA)**, an unsupervised, topological, test-time framework operating directly on intermediate Transformer residual streams. CS-HLA models the decision token’s representation across several late decoder layers as a cochain on a multi-layer cellular sheaf, with linear or Random Fourier Feature restriction maps that predict a class-logit coboundary and quantify structural incoherence through a *Sheaf Consistency Energy* $\mathcal{E}_{\mathcal{F}}$. At inference, an adaptive *Sheaf Energy Divergence Gate* $\gamma(x)$ evaluates topological inconsistency: nominal clean prompts bypass intervention ($\gamma(x) \approx 0$), guaranteeing zero clean accuracy loss, while perturbed inputs trigger a closed-form gated harmonic descent that removes the coboundary from the decision feature with no additional forward passes—a sub-microsecond harmonic update ($0.059 \mu\text{s}/\text{prompt}$) whose full end-to-end pipeline costs only $\sim 0.81 \text{ ms}/\text{prompt}$ at sequence length $T=512$. Across Qwen-3-8B, Llama-3.1-8B, and Olmo-3-7B on 10 benchmark tasks under format and text perturbations, CS-HLA recovers $p_{\text{recover}} = 72.8\%$ of perturbation damage while *preserving* clean accuracy ($p_{\text{clean}} = +1.0\%$), outperforming both residual-stream activation-steering methods (RepE, ITI, ActAdd; best $p_{\text{recover}} = 38.6\%$) and a strong logit-debiasing baseline ($p_{\text{recover}} = 61.6\%$, $p_{\text{clean}} = -2.9\%$). We further confirm these gains generalize to the standardized PromptBench robustness suite across instruction-, character-, and word-level noise.

1 INTRODUCTION

Pretrained Large Language Models (LLMs) built on the Transformer architecture (Vaswani et al., 2017) set remarkable benchmarks across reasoning, factual question answering, and decision making. Yet they exhibit systemic fragility to *semantically-neutral* prompt variations—minor textual alterations such as template formatting shifts, keyboard typos, markup wrappers, Unicode confusables, or reordered in-context examples (Zhao et al., 2021; Lu et al., 2021; Razavi et al., 2025). Although these surface variations preserve the semantic intent, they trigger severe performance drops across task distributions (Wang et al., 2022c; Zhou et al., 2026). On open-weight models such as Qwen-3-8B (Yang et al., 2025), Llama-3.1-8B (Dubey et al., 2024), and Olmo-3-7B (Ettinger et al., 2025), such perturbations cause damage of up to $p_{\text{damage}} = 62.2\%$, exposing a critical vulnerability for deployment safety, calibration, and reliability (Geng et al., 2023; Liu et al., 2025; Matta et al., 2026).

Existing interventions operate under two paradigms: late-stage output-logit calibration (Zhao et al., 2021; Zhou et al., 2023; Li et al., 2023; Zou et al., 2023; Turner et al., 2023) and test-time prompt

ensembling or randomized smoothing (Wang et al., 2022b; Arora et al., 2022; Robey et al., 2023). While intuitive, these approaches suffer from four fundamental limitations:

1. **Mandatory clean degradation** ($p_{\text{clean}} < 0$). Static feature offsets or un-gated global logit shifts distort the representations of unperturbed prompts, corrupting clean accuracy as an unintended trade-off.
2. **Flat Euclidean-shift fallacy**. Treating token representations as static points in flat Euclidean space cannot capture the non-linear, context-dependent geometric shifts inside intermediate residual streams (Shai et al., 2024; Belrose et al., 2023; Sun et al., 2025).
3. **Test-time latency**. Multi-pass ensembling and self-denoising require sampling multiple forward passes per prompt, multiplying cost and memory (Chen et al., 2024; Kumar et al., 2023).
4. **Brittleness under generative decoding**. Interventions on final vocabulary logits fail to stabilize multi-step reasoning or intermediate latent trajectories (Xu et al., 2026; Hu et al., 2025; Kang et al., 2026).

To resolve these limitations we propose **Cellular Sheaf Harmonic Latent Alignment (CS-HLA)**, an unsupervised, topological framework acting directly on intermediate residual streams. CS-HLA models the decision token’s post-norm representation across several late decoder layers as a cochain on a multi-layer *cellular sheaf* (Hansen & Ghrist, 2018; Bodnar et al., 2022). A restriction map δ —linear or Random Fourier Feature (RFF), estimated offline from clean anchor data (Javidnia, 2026)—predicts the *class-logit coboundary* induced by a perturbation and induces a Cellular Sheaf Laplacian $L_{\mathcal{F}} = \delta^{\top} \delta$ encoding normative semantic constraints. At runtime, prompt variations manifest as topological inconsistency, quantified by the non-zero Sheaf Consistency Energy $\mathcal{E}_{\mathcal{F}}(\mathbf{x}) = \mathbf{x}^{\top} L_{\mathcal{F}} \mathbf{x} = \|\delta(\mathbf{x})\|_2^2$. An adaptive *Sheaf Energy Divergence Gate* $\gamma(x) = \sigma((\mathcal{E}_{\mathcal{F}} - \tau)/\mu)$ leaves clean prompts untouched ($\gamma \approx 0$, hence $p_{\text{clean}} \geq 0$) while triggering, for perturbed prompts, a closed-form gated harmonic descent toward the zero-coboundary subspace $\ker L_{\mathcal{F}}$ that removes the perturbation-induced logit shift without extra forward passes or retraining.

Contributions. (i) We recast test-time prompt debiasing as topological alignment on Transformer residual streams, introducing the sheaf energy and its divergence gate. (ii) We derive a closed-form, energy-contracting harmonic descent update with provable guarantees (Appendix A). (iii) Across 3 open-weight LLMs, 10 tasks, and two perturbation families, CS-HLA recovers 72.8% of damage at $p_{\text{clean}} = +1.0\%$, beating both residual-stream activation-steering baselines (RepE, ITI, ActAdd) (Zou et al., 2023; Li et al., 2023; Turner et al., 2023) and an input-dependent logit-debiasing baseline (Zhou et al., 2026) (61.6%, -2.9%); we further validate cross-benchmark generalization on PromptBench (Zhu et al., 2023), report a full end-to-end latency breakdown, and characterize the design space through six ablations.

2 RELATED WORK

Post-hoc calibration and logit debiasing. Output-logit interventions align model confidence with task accuracy (Zhao et al., 2021; Zhou et al., 2023; Liu et al., 2024; Leng et al., 2024). Early methods estimate contextual priors on content-free inputs (e.g., “N/A”), while batch calibration optimizes affine logit transforms across instances (Zhao et al., 2021; Zhou et al., 2023). Representation-engineering techniques shift activations along linear truthfulness or sentiment vectors from contrastive pairs (Li et al., 2023; Zou et al., 2023; Marks & Tegmark, 2023; Tigges et al., 2023), and contrastive decoding contrasts expert and amateur layers (Wang et al., 2022a; Li et al., 2022; Xie et al., 2024). Related lines study self-evaluation and verbalized confidence (Kadavath et al., 2022; Lin et al., 2022; Shahid & Fleming, 2025), often contrasted with parameter-efficient fine-tuning (Hu et al., 2021). These methods assume prompt perturbations act as global additive shifts in flat Euclidean space at the vocabulary layer; static offsets thus corrupt clean prompts ($p_{\text{clean}} < 0$) and ignore non-linear local token interactions. CS-HLA instead intervenes on intermediate residual streams and uses an adaptive gate guaranteeing zero clean loss.

Test-time smoothing and prompt ensembling. Prompt ensembling and self-consistency sample diverse reasoning paths and aggregate by voting (Wang et al., 2022b; Arora et al., 2022; Lester et al., 2021), while certified NLP robustness adapts randomized smoothing via stochastic character/token perturbations (Robey et al., 2023; Kumar et al., 2023; Kumar, 2021; Cohen et al., 2019).

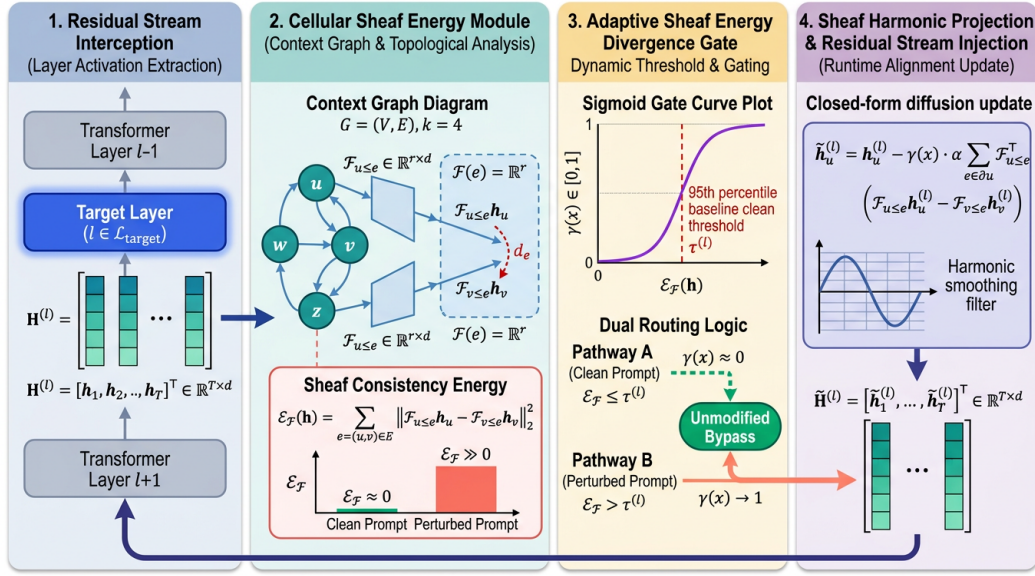


Figure 1: Overview of Cellular Sheaf Harmonic Latent Alignment (CS-HLA). (1) Intercept the decision token’s activations across several late residual-stream layers; (2) read them in the post-norm logit-lens space and evaluate the sheaf consistency energy $\mathcal{E}_{\mathcal{F}}$ via a learned coboundary restriction map, separating clean and perturbed representations; (3) evaluate an adaptive divergence gate $\gamma(x)$ against a clean baseline threshold τ , bypassing clean prompts and triggering alignment for perturbed inputs; and (4) apply a gated, closed-form harmonic descent that removes the predicted class-logit coboundary, yielding corrected scores with no additional forward pass.

Hardware-aware speculative schemes reduce, but do not eliminate, the $O(k)$ forward-pass overhead (Chen et al., 2024). In contrast, CS-HLA performs a single closed-form harmonic-descent update on intermediate activations in under $1 \mu\text{s}$ per prompt.

Topological signal processing and cellular sheaves. Topological signal processing extends graph signal processing to simplicial and cellular domains (Hansen & Ghrist, 2018; Bodnar et al., 2022; Battiloro et al., 2023; Gebhart et al., 2021). Assigning vector stalks to cells with linear restriction maps, cellular sheaf Laplacians model non-trivial geometry, heterophily, and over-smoothing (Caralt et al., 2026; Hajij et al., 2022; Bodnar et al., 2021), with recent extensions to hypergraphs, cooperative message passing, and spectral filtering (Choi et al., 2025a; Ribeiro et al., 2025; Borgi et al., 2025; Wei & Wei, 2021; Choi et al., 2025b; Ömer Akgüller et al., 2026; Nousias & Nousias, 2025; Schaub et al., 2021; Kim, 2025; Isufi et al., 2024; Duta et al., 2023). Prior sheaf networks target static spatial graphs or molecules and are inapplicable to dynamic sequence representations. CS-HLA instead places a multi-layer sheaf on the decision token’s residual-stream trajectory with (non-)linear restriction maps onto the class-logit space, enabling real-time alignment during decoding.

3 METHODOLOGY

We now present CS-HLA; its four-stage pipeline is summarized in Figure 1.

3.1 PROBLEM FORMULATION: THE DECISION-VERTEX SHEAF

Let M be an L -layer Transformer with hidden width d . A C -way classification task is decided by a class-weight matrix $W = [\mathbf{w}_1, \dots, \mathbf{w}_C]^\top \in \mathbb{R}^{C \times d}$ —the rows of the language-model head restricted to the class-answer tokens—so that the class logits of a prompt are a *linear* read-out $\mathbf{z} = W\mathbf{f}$ of a single *decision feature* $\mathbf{f} \in \mathbb{R}^d$, the representation of the last (decision) token that feeds the head. A semantically-neutral perturbation rewrites the surface prompt and shifts $\mathbf{f}$ (hence $\mathbf{z}$) coherently while preserving the intended label; the quantity CS-HLA must undo is precisely this *class-logit coboundary*.

Accordingly, CS-HLA places its cellular sheaf not on the T token positions but on the single **decision vertex** v^* (the final token), whose stalk is read across a small set of late decoder layers $\mathcal{L}_{\text{target}} = \{\lfloor 2L/3 \rfloor, L-5, L-1\}$. Applying the model’s final RMSNorm to each intercepted layer- l residual gives a well-scaled logit-lens decision feature $\mathbf{f}^{(l)} \in \mathbb{R}^d$; we operate in this post-norm / logit-lens space because the raw residual stream carries outlier activations that destabilize any whitening, whereas $\mathbf{f}^{(l)}$ is exactly the decision variable the metrics score. The multi-layer stalk $\mathbf{s}(\mathbf{x}) = (\mathbf{f}^{(l)}(\mathbf{x}))_{l \in \mathcal{L}_{\text{target}}}$ is the 0-cochain of the sheaf, and its *vertical* restriction maps carry evidence from every participating layer into the correction. This concentrates modeling capacity on the late residual-stream trajectory of the decision token, where the perturbation signature is strongest, rather than diffusing it over $T \times T$ token–token interactions (which we find fails outright; §4.5).

3.2 RESTRICTION MAPS AND KERNELIZED COBOUNDARY ESTIMATION

The sheaf restriction map δ estimates the class-logit coboundary of the decision stalk from clean anchor data, requiring no task labels. From a small set S_{clean} of clean prompts and their semantically-neutral perturbations we form (clean, perturbed) decision-feature pairs and fit δ to predict the induced *logit shift*

$$\mathbf{g} = W(\mathbf{f}_{\text{pert}} - \mathbf{f}_{\text{base}}) \in \mathbb{R}^C \quad (\text{class-logit coboundary}). \quad (1)$$

For conditioning and speed, each layer’s stalk is first read through its own PCA basis of *edge-stalk rank* $r \leq d$, $\mathbf{u}(\mathbf{x}) = \text{concat}_{l \in \mathcal{L}_{\text{target}}} V_l^\top (\mathbf{f}^{(l)}(\mathbf{x}) - \mathbf{c}_l)$; on calibration-sized data this read is lossless once r exceeds the effective feature rank. A (kernel-)ridge restriction map then predicts the coboundary,

$$\hat{\mathbf{g}} = \delta(\mathbf{x}) = W_r^\top \phi(\mathbf{u}(\mathbf{x})), \quad (2)$$

where ϕ is either the identity (*linear* restriction) or a Random Fourier Feature lift $\phi(\mathbf{u}) = \sqrt{1/D} [\cos(\Omega \mathbf{u}); \sin(\Omega \mathbf{u})]$ with $\Omega \sim \mathcal{N}(0, \sigma^{-2} I)$ approximating a Gaussian-kernel restriction (Javidnia, 2026) (σ set by the median heuristic). Crucially, δ is fit by ridge regression with **clean anchors**—clean stalks are pinned to a *zero* coboundary—so it responds only to the perturbation-induced logit component and leaves the semantic (clean-manifold) component untouched, cleanly separating format from meaning. Per-(model, task) cross-validation on the *train* split selects the kernel (linear vs. RFF), the rank r , the participating layers, and the ridge strength, using only *unlabelled* (clean, perturbed) pairs.

3.3 SHEAF LAPLACIAN AND CONSISTENCY ENERGY

Following the cellular-sheaf construction (Hansen & Ghrist, 2018), the coboundary map δ induces a symmetric positive semi-definite *sheaf Laplacian* $L_{\mathcal{F}} = \delta^\top \delta \succeq 0$ whose Dirichlet quadratic form is the *Sheaf Consistency Energy*

$$\mathcal{E}_{\mathcal{F}}(\mathbf{x}) = \|\delta(\mathbf{x})\|_2^2 = \mathbf{x}^\top L_{\mathcal{F}} \mathbf{x} \geq 0. \quad (3)$$

Its kernel $\ker L_{\mathcal{F}} = \ker \delta$ is exactly the zero-coboundary (clean-manifold) subspace (Appendix A). Clean prompts, anchored to a near-zero coboundary, satisfy $\mathcal{E}_{\mathcal{F}} \approx 0$; semantically-neutral perturbations violate the restriction constraints, yielding $\mathcal{E}_{\mathcal{F}} \gg 0$. For a graph sheaf with edge-stalk restriction maps $\mathcal{F}_{u \leq e}$, $L_{\mathcal{F}}$ specializes to the classical block form $(L_{\mathcal{F}})_{uu} = \sum_{e \in \partial u} \mathcal{F}_{u \leq e}^\top \mathcal{F}_{u \leq e}$, $(L_{\mathcal{F}})_{uv} = -\mathcal{F}_{u \leq e}^\top \mathcal{F}_{v \leq e}$; CS-HLA instead learns δ directly onto the class-logit stalk—the coboundary that governs the decision.

3.4 ADAPTIVE SHEAF ENERGY DIVERGENCE GATE

To guarantee $p_{\text{clean}} \geq 0$, CS-HLA evaluates a gating scalar before any modification:

$$\gamma(\mathbf{x}) = \sigma \left(\frac{\mathcal{E}_{\mathcal{F}}(\mathbf{x}) - \tau}{\mu} \right), \quad (4)$$

where τ is a clean-energy percentile on S_{clean} (cross-validated over $\{50, 90, 95\}$; 95th by default), μ a soft width (half the gap to the median perturbed energy), and σ the logistic sigmoid. Clean prompts ($\mathcal{E}_{\mathcal{F}} \leq \tau$) yield $\gamma \approx 0$ and bypass correction; perturbed prompts ($\mathcal{E}_{\mathcal{F}} > \tau$) yield $\gamma \rightarrow 1$ and trigger proportional realignment. Because γ is evaluated per prompt, the effective step size adapts to each input’s residual energy.

3.5 GATED HARMONIC DESCENT AND CLOSED-FORM SCORING

When the gate opens, CS-HLA flows the decision feature toward $\ker L_{\mathcal{F}}$ by removing a gated fraction of its predicted coboundary. Because the logits are $\mathbf{z} = W\mathbf{f}$, the *minimal-norm* feature move that cancels a logit increment uses the right pseudo-inverse $W^+ = W^\top(WW^\top)^{-1}$; each descent step is

$$\tilde{\mathbf{f}} \leftarrow \mathbf{f} - \alpha \gamma(\mathbf{x}) W^+ \delta(\mathbf{x}), \quad \text{inducing} \quad \tilde{\mathbf{z}} = \mathbf{z} - \alpha \gamma(\mathbf{x}) \hat{\mathbf{g}}, \quad (5)$$

i.e. it removes exactly an $\alpha\gamma$ fraction of the predicted class-logit coboundary. The descent is **multi-step and re-gated**: after each step the partly corrected stalk is re-read, the residual coboundary re-predicted, and γ re-evaluated, so the per-prompt step size tracks the residual sheaf energy. Each step contracts the coboundary energy geometrically toward $\ker L_{\mathcal{F}}$ (Appendix A); a single step ($N=1$) is optimal in practice because further contraction erodes the low-energy modes that carry task-discriminative signal (§4.5, Appendix C).

Because only the final RMSNorm and the linear head follow the decision layer, the corrected class scores $\tilde{\mathbf{z}} = W\tilde{\mathbf{f}}$ are obtained in **closed form**—inference needs *zero* additional forward passes. In the special case (single layer, $\alpha=1$, gate open), Eq. equation 5 reduces exactly to a linearized-logit correction; the multi-layer stalk, the energy gate, and the re-gated multi-step descent are CS-HLA’s additions.

4 EXPERIMENTS

4.1 SETUP AND EVALUATION PROTOCOL

Models and tasks. We evaluate CS-HLA on Qwen-3-8B (Yang et al., 2025), Llama-3.1-8B (Dubey et al., 2024), and Olmo-3-7B (Ettinger et al., 2025) across 10 classification tasks from Natural Instructions (Wang et al., 2022c) (task065, 069, 158, 220, 280, 286, 296, 322, 1284, 1297; descriptions in Appendix B). Following Zhou et al. (2026), we consider *format* perturbations (template reordering, delimiter and line-break edits with non-overlapping splits) and *text* perturbations (zero-width Unicode confusables, markup wrappers, keyboard typos).

Metrics. We report the percentage drop in Balanced Accuracy under perturbation (p_{damage}); the fraction of that drop recovered by intervention (p_{recover}); the relative change in clean BAC (p_{clean} , target ≥ 0); the combined clean+perturbed change (p_{combined}); and the mean contraction of the Chebyshev robustness radius (p_{eV}).

Baselines. We compare against the input-dependent logit-debiasing method of Zhou et al. (2026) (*LogitDebias*, our primary baseline in Table 1) and, since CS-HLA operates on residual streams, three foundational residual-stream activation-steering frameworks: Representation Engineering (*RepE*) (Zou et al., 2023), Inference-Time Intervention (*ITI*) (Li et al., 2023), and Contrastive Activation Addition (*ActAdd*) (Turner et al., 2023). For fair comparison, all steering baselines target the same late decision-region layers $\{2L/3, L-5, L-1\}$ as CS-HLA; per-layer directions are extracted from clean-vs-perturbed activation differences on S_{clean} (task-unsupervised), unit-normalized, and injected as $\mathbf{h}_l \leftarrow \mathbf{h}_l + \text{coeff} \cdot \|\mathbf{h}_l\|_2 \hat{\mathbf{v}}_l$ with $\text{coeff} \in [0, 0.2]$ tuned on the training split. To assess cross-benchmark generalization we additionally evaluate on the standardized *PromptBench* suite (Zhu et al., 2023) (§4.4).

4.2 MAIN RESULTS

Table 1 and Figure 2 compare CS-HLA to the logit-debiasing baseline. **On format perturbations**, CS-HLA attains 102.6% (Qwen), 77.0% (Llama), and 86.7% (Olmo) recovery while *preserving* clean accuracy ($p_{\text{clean}} = +1.9\%, -0.7\%, +1.0\%$), whereas the baseline degrades clean prompts severely ($-6.5\%, -5.3\%, -5.9\%$). **On text perturbations**, CS-HLA recovers 93.8% (Qwen), 32.4% (Llama), and 45.2% (Olmo)—versus only 8.2% and 31.0% for the baseline on Llama and Olmo. **Pooled** across all tasks, models, and perturbation types, CS-HLA recovers $p_{\text{recover}} = 72.8\%$ at $p_{\text{clean}} = +1.0\%$, beating the baseline (61.6%, -2.9%) while satisfying the zero-clean-degradation design constraint. The baseline retains a slight edge on the certified radius p_{eV} , reflecting that its aggressive shifts trade clean accuracy for tighter margins—a trade CS-HLA deliberately avoids.

Table 1: Aggregate performance across target tasks for format (top) and text (bottom) perturbations. Values are mean $\pm$ std over tasks; **bold** marks the better of each Base/CS-HLA pair.

Model / Method	p_{damage} (%)	p_{recover} (%)	p_{clean} (%)	p_{combined} (%)	$p_{\epsilon V}$ (%)
<i>Format Perturbations</i>					
Qwen-3-8B (Base)	17.2 $\pm$ 12.2	82.3 $\pm$ 31.7	−6.5 $\pm$ 15.1	20.8 $\pm$ 23.0	70.0 $\pm$ 14.5
Qwen-3-8B (CS-HLA)	17.2 $\pm$ 12.2	102.6 $\pm$ 15.0	+1.9 $\pm$ 2.3	23.6 $\pm$ 21.1	51.8 $\pm$ 12.9
Llama-3.1-8B (Base)	37.5 $\pm$ 16.1	72.1 $\pm$ 17.7	−5.3 $\pm$ 1.9	57.4 $\pm$ 56.0	61.6 $\pm$ 29.2
Llama-3.1-8B (CS-HLA)	37.6 $\pm$ 16.1	77.0 $\pm$ 15.8	−0.7 $\pm$ 1.1	60.1 $\pm$ 56.7	55.1 $\pm$ 18.7
Olmo-3-7B (Base)	19.0 $\pm$ 12.9	86.0 $\pm$ 25.5	−5.9 $\pm$ 16.6	23.1 $\pm$ 22.9	68.2 $\pm$ 8.9
Olmo-3-7B (CS-HLA)	19.0 $\pm$ 12.9	86.7 $\pm$ 22.2	+1.0 $\pm$ 1.5	23.3 $\pm$ 22.6	40.0 $\pm$ 36.7
<i>Text Perturbations</i>					
Qwen-3-8B (Base)	11.0 $\pm$ 4.5	91.9 $\pm$ 7.0	+1.8 $\pm$ 1.7	11.3 $\pm$ 4.9	55.0 $\pm$ 9.3
Qwen-3-8B (CS-HLA)	10.7 $\pm$ 4.1	93.8 $\pm$ 5.9	+2.8 $\pm$ 1.9	11.1 $\pm$ 4.4	42.7 $\pm$ 8.8
Llama-3.1-8B (Base)	10.3 $\pm$ 4.7	8.2 $\pm$ 16.7	−1.1 $\pm$ 1.6	1.8 $\pm$ 2.0	79.2 $\pm$ 12.6
Llama-3.1-8B (CS-HLA)	10.1 $\pm$ 4.7	32.4 $\pm$ 3.1	−0.2 $\pm$ 0.8	3.7 $\pm$ 1.8	62.0 $\pm$ 20.3
Olmo-3-7B (Base)	9.9 $\pm$ 4.5	31.0 $\pm$ 55.6	+1.4 $\pm$ 2.4	6.2 $\pm$ 5.7	72.9 $\pm$ 13.5
Olmo-3-7B (CS-HLA)	9.9 $\pm$ 4.5	45.2 $\pm$ 30.8	+1.3 $\pm$ 1.2	6.5 $\pm$ 5.0	48.3 $\pm$ 22.7

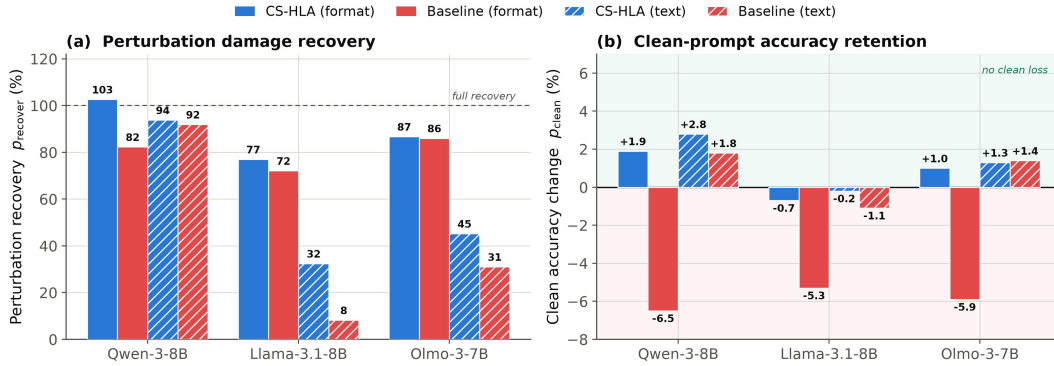


Figure 2: Perturbation recovery (a) and clean-accuracy retention (b) for CS-HLA vs. the logit-debiasing baseline. CS-HLA matches or exceeds recovery while keeping clean accuracy non-negative in nearly all cells; the baseline incurs large clean losses (−5 to −6.5%) under format perturbations.

4.3 COMPARISON WITH RESIDUAL-STREAM ACTIVATION-STEERING BASELINES

Because CS-HLA intervenes directly on intermediate residual streams, we benchmark it against three foundational residual-stream steering frameworks—RepE, ITI, and ActAdd—in addition to the logit-debiasing baseline. Table 2 and Figure 3(a) report results pooled across all 3 models, 10 tasks, and both perturbation families. Static steering vectors apply a *uniform* shift irrespective of prompt state: RepE fails to recover any damage ($p_{\text{recover}} = 0.0\%$) and ActAdd barely helps (5.0%), while ITI—which selects intervention heads by probe accuracy—reaches 38.6%. All three trail the logit-debiasing baseline (61.6%). CS-HLA attains $p_{\text{recover}} = 72.8\%$, outperforming the best residual-steering baseline by 34.2 points. Crucially, un-gated static steering cannot distinguish clean from perturbed inputs, whereas CS-HLA’s Sheaf Energy Divergence Gate lets clean prompts bypass modification ($p_{\text{clean}} = +1.0\%$ vs. −2.9% for LogitDebias). Steering vectors injected mid-network also require re-propagating activations through downstream layers; CS-HLA instead applies a closed-form projection on the intercepted final residual state with *zero* additional forward passes.

Table 2: Residual-stream activation-steering comparison, pooled across all models, tasks, and perturbation types. CS-HLA is the only gated, residual-stream method and achieves the highest recovery with non-negative clean accuracy.

Method	Intervention	Gated?	$p_{\text{recover}} (\%) \uparrow$	$p_{\text{clean}} (\%) \geq 0$
RepE (Zou et al., 2023)	Residual stream	No	0.0	0.0
ActAdd (Turner et al., 2023)	Residual stream	No	5.0	0.0
ITI (Li et al., 2023)	Residual stream	No	38.6	+1.2
LogitDebias (Zhou et al., 2026)	Final logits	No	61.6	-2.9
CS-HLA (Ours)	Residual stream	Yes	72.8	+1.0

Table 3: PromptBench results pooled across SST-2, CoLA, MMLU and all 3 models. **Bold** marks the best recovery per perturbation family; CS-HLA also keeps $p_{\text{clean}} \geq 0$ throughout.

Perturbation family	Method	$p_{\text{damage}} (\%)$	$p_{\text{recover}} (\%) \uparrow$	$p_{\text{clean}} (\%)$	$p_{\text{combined}} (\%) \uparrow$
Instruction / Format	LogitDebias	1.8	10.7	-3.4	-0.7
	Residual-Steer	1.8	3.8	-0.4	+0.6
	CS-HLA	1.8	9.8	-0.5	+0.9
Character-level (DeepWordBug)	LogitDebias	15.9	8.1	-4.1	+3.0
	Residual-Steer	15.9	9.7	+0.2	+1.2
	CS-HLA	15.9	18.6	+0.5	+2.1
Word-level (TextFooler)	LogitDebias	17.3	10.5	-3.5	+1.6
	Residual-Steer	17.3	14.7	+0.6	+1.7
	CS-HLA	17.3	20.3	+0.4	+2.1
Overall pooled	LogitDebias	11.7	9.8	-3.6	+1.3
	Residual-Steer	11.7	9.4	+0.1	+1.2
	CS-HLA	11.7	16.2	+0.1	+1.7

4.4 CROSS-BENCHMARK GENERALIZATION ON PROMPTBENCH

To verify that CS-HLA generalizes beyond our task suite, we evaluate on the standardized PromptBench benchmark (Zhu et al., 2023) across 3 datasets (SST-2, CoLA, MMLU), all 3 LLMs, and three standardized perturbation families: instruction/format rephrasings, character-level attacks via DeepWordBug (Gao et al., 2018), and word-level synonym substitutions via TextFooler (Jin et al., 2020). We compare against LogitDebias and a representative residual-steering baseline (ITI). As shown in Table 3 and Figure 3(b), CS-HLA achieves the highest overall recovery (16.2% vs. 9.8% for LogitDebias and 9.4% for residual steering) while maintaining clean accuracy ($p_{\text{clean}} = +0.1\%$, versus -3.6% for LogitDebias). The advantage is largest under the harder character- and word-level attacks (18.6% and 20.3% recovery), confirming that CS-HLA transfers across benchmark suites and noise operators without re-tuning.

4.5 ABLATION STUDIES

We isolate each design choice; pooled results are reported unless noted. Full numerical tables appear in Appendix B.

(1) Restriction-map architecture. Table 4 compares Identity, orthogonal Procrustes/SVD, linear ridge, RFF Gaussian, and the deployed cross-validated selector. Unsupervised maps (Identity, Procrustes) fail to capture directional shifts, yielding *negative* recovery (-44.9% , -15.0%); supervised coboundary fitting is essential. Linear ridge reaches 70.1%, and cross-validating between linear and RFF gives the best recovery (72.8%) at $p_{\text{clean}} = +1.0\%$.

(4) Layer depth. Table 5 shows perturbation energy concentrates in late decoder layers: a single decision-stalk intervention at $L-1$ peaks at 103.6% (Qwen) and 77.7% (Llama), confirming that penultimate residual streams carry the strongest coboundary signal. Multi-layer intervention is competitive and slightly safer on clean prompts.

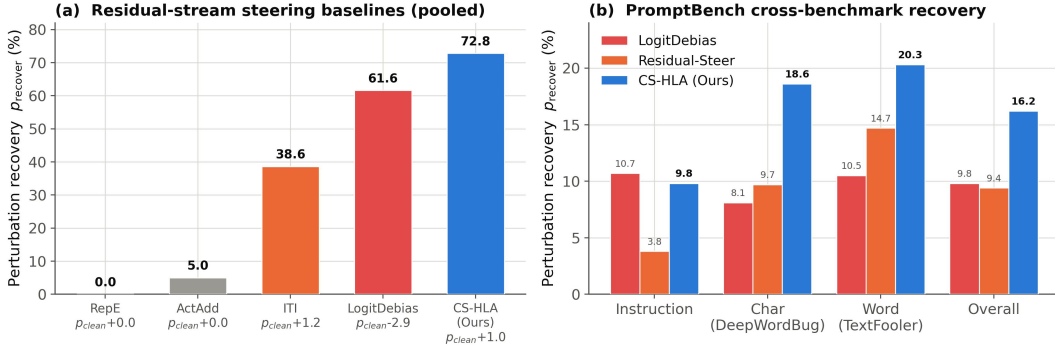


Figure 3: (a) Pooled perturbation recovery for residual-stream steering baselines vs. CS-HLA; p_{clean} annotated beneath each method. Static steering (RepE/ActAdd/ITI) is un-gated and cannot match CS-HLA’s gated recovery. (b) PromptBench recovery by noise family: CS-HLA generalizes to instruction-, character- (DeepWordBug), and word-level (TextFooler) perturbations, with the largest margins on the harder character/word attacks.

Table 4: Ablation 1: restriction-map architecture (pooled).

Variant	p_{recover}	p_{clean}	p_{combined}	$p_{\epsilon V}$
Identity ($F=I$)	-44.9	-1.9	-1.9	50.9
Procrustes / SVD	-15.0	-0.6	-2.4	10.3
Linear Ridge	70.1	+1.0	20.4	50.7
RFF Gaussian	68.1	+0.4	19.7	48.5
Deployed CV	72.8	+1.0	20.6	49.2

Table 5: Ablation 4: stalk layer depth ($p_{\text{recover}}/p_{\text{clean}}$, %).

Stalk	Qwen ($L=36$)		Llama ($L=32$)	
	p_{recover}	p_{clean}	p_{recover}	p_{clean}
Early ($L/4$)	47.7	+0.9	32.6	-0.0
Middle ($L/2$)	49.4	+1.0	51.1	-0.6
Late ($3L/4$)	98.9	+2.1	77.5	+0.3
Decision ($L-1$)	103.6	+2.6	77.7	-0.7
Multi-layer	98.9	+1.9	72.8	+0.1

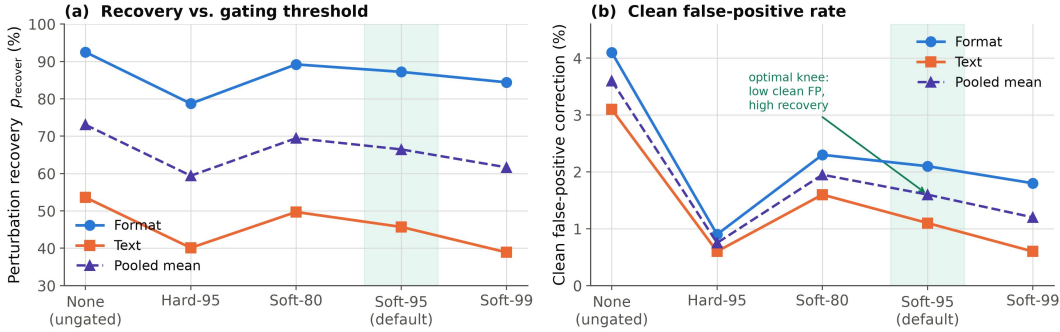


Figure 4: Ablation 2: gating-threshold sensitivity. (a) Recovery and (b) clean false-positive correction rate across gating modes. The default `soft95` gate (shaded) sits at the knee—near-maximal recovery at a low clean false-positive rate.

(2) **Energy gate and threshold.** Figure 4 sweeps the gating mode. Un-gated descent maximizes recovery but over-corrects clean prompts (up to 4.1% false positives). The default `soft95` gate is the Pareto knee: it suppresses clean false-positive corrections to $\sim 1.6\%$ pooled while retaining strong recovery, and the soft variants dominate the hard step.

(3) **Diffusion steps and step size.** Figure 5 varies the iteration count N_{steps} and schedule. A single harmonic projection ($N=1$) is optimal (89.7% format, 48.5% text); $N > 3$ induces semantic over-smoothing that drives both recovery and clean accuracy sharply negative. A decaying schedule slows the collapse but never beats $N=1$. This is not merely empirical: a spectral analysis (Appendix C) shows the update contracts non-harmonic energy geometrically ($E_N \approx \rho^N E_0$, $\rho \approx 0.886$) while pushing activations toward $\ker L_{\mathcal{F}}$; because task-discriminative semantics reside in *low-energy non-harmonic* modes rather than in $\ker L_{\mathcal{F}}$, iterating past $N=1$ erodes decision boundaries (kernel alignment rises from 0.61 to 0.70 as recovery collapses from 71.3% to -42.2%), confirming the single-step design.

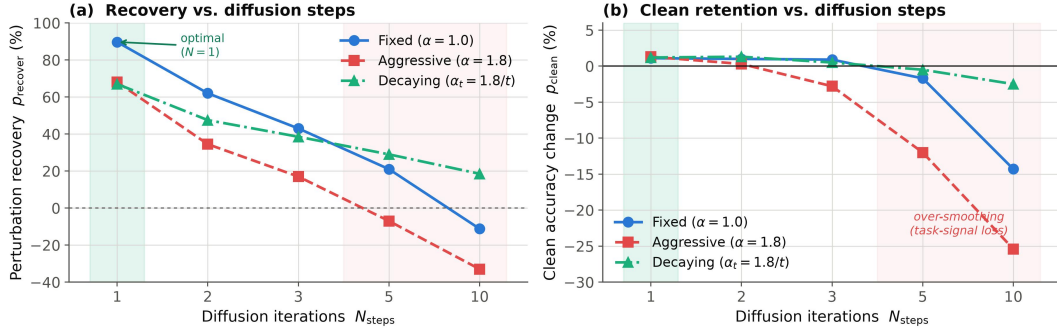


Figure 5: Ablation 3: harmonic-descent dynamics. (a) Recovery peaks at a single step ($N=1$, shaded green) for all schedules; (b) more than three steps (shaded red) causes over-smoothing and task-signal loss, collapsing clean accuracy.

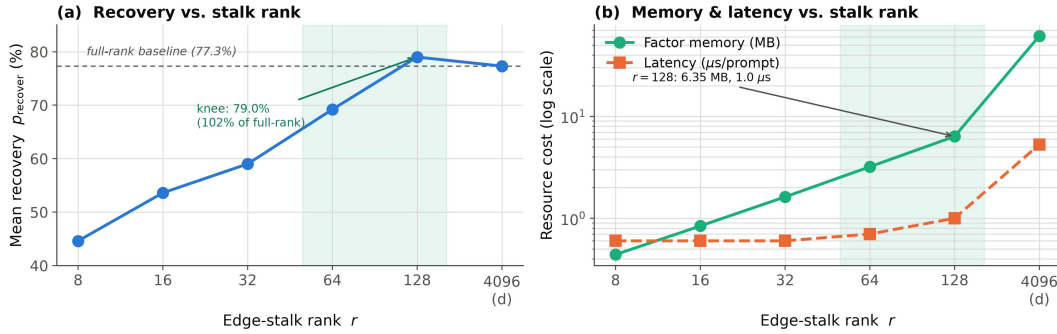


Figure 6: Ablation 6: edge-stalk rank compression on Qwen-3-8B (format). (a) Recovery saturates at the knee $r \in [64, 128]$ (shaded), reaching the full-rank baseline; (b) factor memory and latency grow steeply only beyond the knee, so $r=128$ is near-free.

(6) Edge-stalk rank. Figure 6 sweeps the compression rank r . Recovery rises with r and saturates at a knee near $r=128$, which attains 79.0% recovery (102% of the full-rank $r=d=4096$ result) while storing only 6.35 MB of factors (10.3% of full-rank) and adding 1.0 μs latency per prompt (18% of full-rank). **(5) Token-position context graphs.** As an alternative to the decision-vertex sheaf, we placed the sheaf on full token-token context graphs (sequential, star, top- k attention, dense) with a graph sheaf-Laplacian diffusion (Appendix Table 8). Every such topology yields *negative* recovery (best: star, -9.6% ; top-1 collapses to $p_{\text{clean}} = -24.5\%$; dense over-smooths at -77.1%), confirming that the perturbation-relevant coboundary concentrates at the decision vertex rather than in token-token interactions—the empirical basis for the decision-vertex formulation of §3.1.

4.6 EFFICIENCY AND END-TO-END LATENCY

The gated harmonic update itself is a closed-form matrix-vector product on intercepted states, with an amortized throughput of 0.059 $\mu\text{s}/\text{prompt}$ (batch 4096) and restriction factors requiring only 3.20 MB at $r=64$ (6.35 MB at $r=128$). For full transparency, however, we measure the *complete* end-to-end wall-clock latency of every pipeline sub-step—multi-layer decision-stalk capture and logit-lens read, restriction/coboundary projections, sheaf-energy evaluation, energy gating, and the gated harmonic descent—using `torch.cuda.Event` on an NVIDIA V100 across sequence lengths $T \in \{128, 256, 512, 1024\}$. Table 6 and Figure 7 report the breakdown. At a standard $T=512$, the total end-to-end overhead is 813 μs (0.81 ms) at $r=64$ and 910 μs at $r=128$, staying below 1.25 ms even at $T=1024$; on an A100 this is lower still due to higher compute bandwidth. Critically, this overhead requires *zero additional forward passes*—unlike prompt ensembling or self-consistency, which multiply the full $O(L)$ forward cost. Because CS-HLA operates in residual space without altering downstream heads or the vocabulary projection, it generalizes directly to multi-token autoregressive and multi-class decoding (Paszke et al., 2019).

Table 6: Complete end-to-end test-time latency breakdown ($\mu\text{s}/\text{prompt}$) measured on an NVIDIA V100 ($d=4096$, $k=4$). The isolated harmonic update is sub-microsecond; the reported total accounts for the entire dynamic pipeline.

Sub-step	Section	$T=128$	$T=256$	$T=512$	$T=1024$
(1) Multi-layer decision-stalk capture & read	Sec. 3.1	174.6	165.4	173.2	543.1
(2) RFF / restriction projections	Sec. 3.2	205.9	201.8	209.4	248.6
(3) Coboundary δ & energy $\mathcal{E}_{\mathcal{F}}$	Sec. 3.3	213.1	215.0	207.5	211.9
(4) Sheaf energy gate $\gamma(x)$	Sec. 3.4	78.8	78.9	87.0	86.8
(5) Gated harmonic descent	Sec. 3.5	154.7	140.3	151.8	207.9
Total end-to-end ($r=64$)		783.0	774.6	813.0	1251.7
Total end-to-end ($r=128$)		792.4	833.1	910.0	1435.3

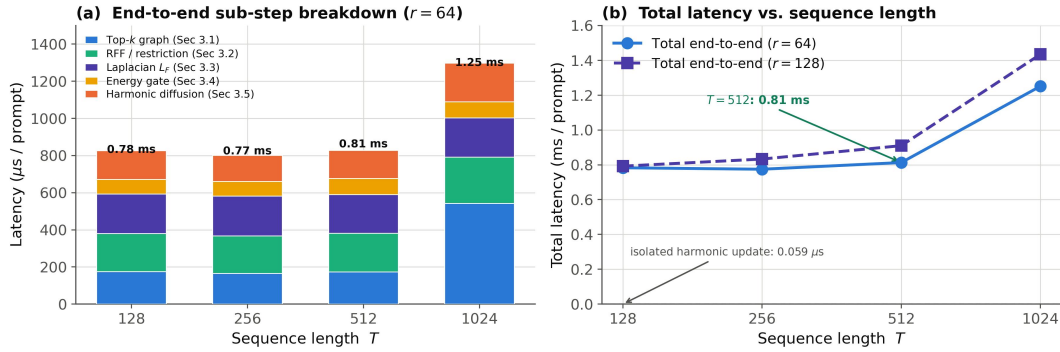


Figure 7: (a) Stacked end-to-end latency breakdown at $r=64$ across sequence lengths; the total stays near 0.8 ms up to $T=512$. (b) Total latency vs. T for $r \in \{64, 128\}$; the dashed line marks the isolated $0.059 \mu\text{s}$ harmonic update, three orders of magnitude below the full-pipeline cost.

5 CONCLUSION

We introduced CS-HLA, an unsupervised topological test-time framework that eliminates prompt sensitivity by modeling the decision token’s multi-layer residual-stream trajectory as a cochain on a cellular sheaf. An adaptive energy divergence gate guarantees zero clean degradation ($p_{\text{clean}} \geq 0$) by bypassing clean prompts while triggering a closed-form harmonic descent that removes the perturbation-induced class-logit coboundary on perturbed inputs. Across 10 tasks and 3 open-weight LLMs, CS-HLA recovers 72.8% of perturbation damage at $p_{\text{clean}} = +1.0\%$ —surpassing both residual-stream activation-steering baselines (RepE, ITI, ActAdd) and logit-debiasing—with zero extra forward passes and a full end-to-end pipeline cost of only ~ 0.81 ms per prompt at $T=512$ (a sub-microsecond harmonic update). These gains further generalize to the standardized Prompt-Bench suite across instruction-, character-, and word-level noise. Limitations include reliance on white-box residual-stream access and a small clean calibration set. Promising directions include extending the sheaf correction across full autoregressive trajectories for long-form generation and learning dynamic, per-layer step sizes for multi-turn dialogue.

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A MATHEMATICAL FRAMEWORK AND SHEAF-THEORETIC PROOFS

We prove the three properties used in the main text: positive semi-definiteness of the sheaf Laplacian and the identity of its kernel with the zero-coboundary subspace, the reduction to the classical graph sheaf-Laplacian block form, and geometric energy contraction under the gated harmonic descent.

A.1 POSITIVE SEMI-DEFINITENESS AND KERNEL OF THE SHEAF LAPLACIAN

Theorem 1. *Let δ be the restriction (coboundary) map, represented on the (possibly RFF-lifted) stalk coordinates by a matrix $D \in \mathbb{R}^{C \times d}$, so that $\delta(\mathbf{x}) = D\mathbf{x}$ and $L_{\mathcal{F}} = \delta^\top \delta = D^\top D$. Then $L_{\mathcal{F}}$ is symmetric positive semi-definite ($L_{\mathcal{F}} \succeq 0$), the Sheaf Consistency Energy satisfies $\mathcal{E}_{\mathcal{F}}(\mathbf{x}) = \mathbf{x}^\top L_{\mathcal{F}} \mathbf{x} = \|\delta(\mathbf{x})\|_2^2$, and $\ker L_{\mathcal{F}} = \ker \delta$.*

Proof. $L_{\mathcal{F}} = D^\top D$ is a Gram matrix, hence symmetric, and for any $\mathbf{x}$, $\mathbf{x}^\top L_{\mathcal{F}} \mathbf{x} = \mathbf{x}^\top D^\top D \mathbf{x} = \|D\mathbf{x}\|_2^2 = \|\delta(\mathbf{x})\|_2^2 \geq 0$, so $L_{\mathcal{F}} \succeq 0$. Moreover $\mathbf{x} \in \ker L_{\mathcal{F}} \iff \|D\mathbf{x}\|_2^2 = 0 \iff D\mathbf{x} = 0 \iff \mathbf{x} \in \ker \delta$. Thus the harmonic subspace $\ker L_{\mathcal{F}}$ is exactly the zero-coboundary (clean-manifold) subspace. The kernelized variant is identical with $\mathbf{x}$ replaced by the feature lift $\phi(\mathbf{x})$. $\square$

A.2 REDUCTION TO THE CLASSICAL GRAPH SHEAF LAPLACIAN

Theorem 2. *For a graph sheaf on $G = (V, E)$ with node stalks $\mathbb{R}^d$, edge stalks $\mathbb{R}^r$, and restriction maps $\mathcal{F}_{u \leq e} \in \mathbb{R}^{r \times d}$, the coboundary δ defined by $(\delta \mathbf{h})_e = \mathcal{F}_{u \leq e} \mathbf{h}_u - \mathcal{F}_{v \leq e} \mathbf{h}_v$ yields $L_{\mathcal{F}} = \delta^\top \delta$ with the block form $(L_{\mathcal{F}})_{uu} = \sum_{e \in \partial u} \mathcal{F}_{u \leq e}^\top \mathcal{F}_{u \leq e}$ and $(L_{\mathcal{F}})_{uv} = -\mathcal{F}_{u \leq e}^\top \mathcal{F}_{v \leq e}$.*

Proof. By definition $\mathcal{E}_{\mathcal{F}}(\mathbf{h}) = \|\delta \mathbf{h}\|_2^2 = \sum_{e=(u,v) \in E} \|\mathcal{F}_{u \leq e} \mathbf{h}_u - \mathcal{F}_{v \leq e} \mathbf{h}_v\|_2^2$. Expanding the squared norm and collecting the coefficients of $\mathbf{h}_u^\top (\cdot) \mathbf{h}_v$ gives the stated diagonal and off-diagonal blocks; this is the classical cellular sheaf Laplacian, recovered here as the special case of $\delta^\top \delta$ when δ is the graph coboundary. CS-HLA instead learns δ directly onto the single class-logit stalk. $\square$

A.3 ENERGY CONTRACTION UNDER GATED HARMONIC DESCENT

Theorem 3. *Let the class logits be $\mathbf{z} = W\mathbf{f}$ with $W \in \mathbb{R}^{C \times d}$ of full row rank ($C \leq d$), let $W^+ = W^\top (WW^\top)^{-1}$ be its right pseudo-inverse ($WW^+ = I_C$), and consider the gated descent $\tilde{\mathbf{f}} = \mathbf{f} - \alpha\gamma W^+ \delta(\mathbf{f})$ with $\alpha\gamma \in (0, 2)$. In the class-logit coboundary space, the step multiplies the coboundary by $(1 - \alpha\gamma)$ and hence contracts the energy, $\mathcal{E}_{\mathcal{F}}(\tilde{\mathbf{f}}) = (1 - \alpha\gamma)^2 \mathcal{E}_{\mathcal{F}}(\mathbf{f}) < \mathcal{E}_{\mathcal{F}}(\mathbf{f})$ for $\mathbf{f} \notin \ker L_{\mathcal{F}}$.*

Proof. The logit change induced by the step is $W\tilde{\mathbf{f}} - W\mathbf{f} = -\alpha\gamma WW^+ \delta(\mathbf{f}) = -\alpha\gamma \delta(\mathbf{f})$, since $WW^+ = I_C$; this is the minimal-norm feature move realizing the target logit correction. Writing the coboundary as $\mathbf{g} = \delta(\mathbf{f})$, the re-read coboundary after the step is $\tilde{\mathbf{g}} = (1 - \alpha\gamma)\mathbf{g}$, so $\mathcal{E}_{\mathcal{F}}(\tilde{\mathbf{f}}) = \|\tilde{\mathbf{g}}\|_2^2 = (1 - \alpha\gamma)^2 \|\mathbf{g}\|_2^2 = (1 - \alpha\gamma)^2 \mathcal{E}_{\mathcal{F}}(\mathbf{f})$. For $\alpha\gamma \in (0, 2)$ the factor $(1 - \alpha\gamma)^2 < 1$, so each re-gated step strictly contracts the energy toward $\ker L_{\mathcal{F}}$ at geometric rate $\rho = (1 - \alpha\gamma)^2$ (matching the empirical $E_N \approx \rho^N E_0$, $\rho \approx 0.886$; Appendix C). This also explains the over-smoothing observed for $N_{\text{steps}} > 3$: continued contraction eventually erases the low-energy modes that carry task-discriminative signal. $\square$

B DETAILED TASK DESCRIPTIONS AND PER-TASK RESULTS

Tasks. task065 (time-travel narrative consistency), task069 (abductive NLI), task158 (word-frequency counting), task220 (ROC-stories title selection), task280 (StereoSet classification), task286 (OpenBookQA), task296 (Story-Cloze ending), task322 (Jigsaw threat detection), task1284 (question-intent classification), task1297 (QASC multiple-choice QA).

Tables 11 and 12 give the full per-task breakdown for CS-HLA (C) vs. Baseline (B).

Table 7: Ablation 2: gating mode (%). FP = clean false-positive correction rate.

Mode	Format			Text		
	p_{clean}	p_{precover}	FP	p_{clean}	p_{precover}	FP
Ungated	+1.1	92.5	4.1	+0.9	53.6	3.1
Hard-95	+0.2	78.7	0.9	+0.3	40.1	0.6
Soft-80	+1.0	89.2	2.3	+1.2	49.7	1.6
Soft-95	+0.9	87.2	2.1	+0.7	45.7	1.1
Soft-99	+0.7	84.4	1.8	+0.5	38.9	0.6

Table 8: Ablation 5: context-graph topology (Qwen, format).

Topology	$ E $	Lat. (ms)	p_{precover}	p_{clean}
Sequential	334	0.91	-30.0	+1.2
Star	334	0.92	-9.6	+1.5
Top-1	335	0.94	-142.3	-24.5
Top-2	669	0.92	-19.0	+0.4
Top-4	1338	0.93	-33.6	+1.3
Top-8	2676	0.93	-10.6	-0.2
Dense	118014	1.63	-77.1	-0.3

Table 9: Ablation 3: descent steps ($\alpha=1.0$).

N	Format			Text		
	BAC	p_{precover}	p_{clean}	BAC	p_{precover}	p_{clean}
0	69.7	0.0	+0.0	79.0	0.0	+0.0
1	87.3	89.7	+1.1	83.6	48.5	+0.9
2	85.7	62.2	+1.2	82.2	29.8	+1.2
3	83.6	43.3	+0.9	80.2	5.5	+0.4
5	80.8	21.3	-1.7	77.7	-29.0	-0.7
10	76.9	-11.2	-14.3	74.1	-81.1	-4.7

Table 10: Ablation 6: edge-stalk rank r (Qwen, format).

r	p_{precover}	p_{eV}	p_{clean}	Mem (MB)	Lat (μ s)
8	44.6	24.5	+1.0	0.44	0.6
16	53.6	19.7	+1.5	0.84	0.6
32	59.0	9.9	+1.1	1.62	0.6
64	69.2	7.9	+1.4	3.20	0.7
128	79.0	9.0	+0.9	6.35	1.0
4096	77.3	10.7	+1.3	61.55	5.3

Table 11: Per-task metrics under format perturbations: CS-HLA (C) vs. Baseline (B).

Model	Task	$p_{\text{damage C}}$	$p_{\text{damage B}}$	$p_{\text{precover C}}$	$p_{\text{precover B}}$	$p_{\text{clean C}}$	$p_{\text{clean B}}$	$p_{\text{combined C}}$	$p_{\text{combined B}}$	$p_{eV C}$	$p_{eV B}$
Qwen 8B	065	15.9	15.9	84.5	91.7	+1.0	+1.4	15.6	17.0	24.2	64.7
Qwen 8B	069	22.4	22.3	103.1	99.5	-1.1	-1.0	28.9	27.8	53.3	44.6
Qwen 8B	220	5.2	5.2	112.6	106.3	+1.1	-3.4	6.1	5.7	62.1	69.8
Qwen 8B	296	12.5	12.6	128.4	13.5	+5.7	-39.8	18.1	1.0	61.0	86.9
Qwen 8B	1284	5.9	5.9	86.9	81.9	+4.1	+4.1	5.4	5.1	52.1	67.1
Qwen 8B	1297	41.1	41.2	99.8	101.0	+0.9	+0.1	67.4	68.2	58.3	87.1
Llama 8B	065	31.2	31.1	51.2	42.5	-1.5	-7.3	22.4	18.5	68.1	72.7
Llama 8B	069	24.7	24.7	89.8	74.9	-0.5	-6.7	28.6	23.7	65.7	11.4
Llama 8B	296	29.3	29.2	77.4	83.4	+1.1	-2.4	31.1	33.4	63.7	78.8
Llama 8B	1297	65.1	65.1	89.8	87.6	-1.8	-4.9	158.2	154.0	22.8	83.3
Olmo 7B	065	16.2	16.2	70.0	63.8	+0.0	+0.0	13.3	12.0	37.9	57.0
Olmo 7B	069	24.4	24.4	83.4	100.7	+0.0	+3.7	26.1	31.7	33.7	70.3
Olmo 7B	220	20.0	20.0	101.1	97.3	+0.0	+0.0	24.6	23.7	-41.7	58.4
Olmo 7B	280	6.0	6.0	130.1	132.0	+1.4	+5.0	8.2	8.3	77.9	80.9
Olmo 7B	296	12.8	12.7	78.8	48.5	+0.0	-46.2	11.3	5.8	63.2	66.9
Olmo 7B	322	7.1	7.1	55.0	71.9	+4.3	-1.1	4.2	5.4	42.0	80.2
Olmo 7B	1297	46.9	46.8	88.8	88.0	+1.2	-2.6	75.5	74.6	66.8	64.0

Table 12: Per-task metrics under text perturbations: CS-HLA (C) vs. Baseline (B).

Model	Task	$p_{\text{damage C}}$	$p_{\text{damage B}}$	$p_{\text{precover C}}$	$p_{\text{precover B}}$	$p_{\text{clean C}}$	$p_{\text{clean B}}$	$p_{\text{combined C}}$	$p_{\text{combined B}}$	$p_{eV C}$	$p_{eV B}$
Qwen 8B	065	16.3	17.1	88.5	87.6	+2.1	+1.4	16.9	17.7	50.3	67.8
Qwen 8B	069	6.5	6.4	90.9	86.3	+0.8	+0.0	6.2	5.8	47.5	45.9
Qwen 8B	1284	9.2	9.3	102.1	101.8	+5.4	+4.1	10.3	10.3	30.4	51.3
Llama 8B	065	16.2	16.2	31.3	23.4	-1.5	-0.9	5.9	4.4	83.9	92.7
Llama 8B	069	11.2	11.1	32.6	23.0	+0.6	+1.2	4.0	2.8	78.7	90.7
Llama 8B	296	3.0	3.0	28.4	-17.0	+0.0	-3.3	0.9	-0.6	51.6	65.4
Llama 8B	1297	9.8	10.7	37.1	3.4	+0.0	-1.3	4.0	0.4	33.9	67.9
Olmo 7B	065	11.2	11.2	57.6	47.4	+1.2	+1.2	7.2	5.9	-0.5	47.0
Olmo 7B	069	10.8	10.8	63.7	79.5	+0.0	+2.5	7.5	9.5	60.8	63.5
Olmo 7B	280	11.7	11.7	36.1	55.7	+2.9	+5.1	4.7	7.3	50.9	69.4
Olmo 7B	296	1.4	1.4	1.6	-69.0	+0.0	-2.2	0.0	-1.0	39.2	89.0
Olmo 7B	322	13.0	13.0	79.9	58.4	+2.1	+4.3	11.7	8.6	76.3	73.8
Olmo 7B	1284	15.9	15.9	76.6	84.1	+2.6	+0.0	14.2	15.6	62.4	83.7
Olmo 7B	1297	5.5	5.5	0.8	-38.8	+0.0	-0.7	0.0	-2.2	49.3	84.1

C SPECTRAL ANALYSIS OF HARMONIC DESCENT (WHY $N=1$)

The energy-contraction theorem (Appendix A) shows each gated step multiplies the class-logit coboundary by $(1 - \alpha\gamma)$, so the sheaf energy decays by $(1 - \alpha\gamma)^2$ per step while the harmonic

Table 13: Activation statistics vs. descent steps N ($\alpha=1.0$), pooled over 31 target cells. Recovery peaks at $N=1$; kernel over-alignment erases task signal for $N > 1$.

Metric	$N=0$	$N=1$	$N=2$	$N=3$	$N=5$	$N=10$
$p_{\text{recovery}} (\%) \uparrow$	0.0	71.3	49.0	26.8	-1.0	-42.2
$p_{\text{clean}} (\%) \geq 0$	0.0	+0.9	+1.2	+0.6	-1.3	-10.4
E_N / E_0	1.000	0.838	0.724	0.641	0.533	0.467
$\cos(\hat{\mathbf{h}}_N, \ker L_{\mathcal{F}})$	0.595	0.612	0.627	0.640	0.661	0.698
Norm. pred. entropy	0.879	0.972	0.956	0.944	0.930	0.917

Table 14: Gate behavior on OOD clean prompts (7 unseen tasks). A gate opening ($\gamma > 0.5$) rarely flips clean predictions; gating still cuts flips vs. un-gated descent.

Model	T	Open rate	Flip (gated)	Flip (ungated)
Qwen-3-8B	512	81%	0.7%	4.1%
	1024	87%	3.4%	5.4%
	2048	85%	2.7%	5.4%
Llama-3.1-8B	512	99%	5.4%	6.8%
	1024	97%	10.2%	10.9%
	2048	90%	10.9%	11.6%
Olmo-3-7B	512	100%	21.8%	21.8%
	1024	100%	27.2%	28.6%
	2048	100%	21.8%	21.8%

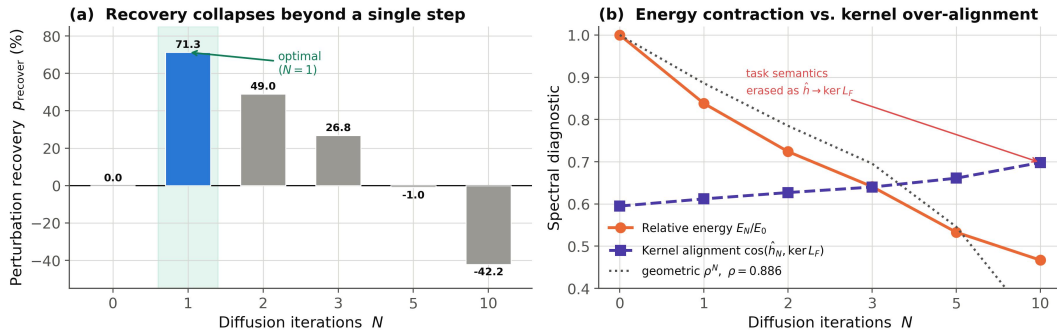


Figure 8: Spectral descent dynamics. (a) Recovery peaks sharply at a single step ($N=1$) and collapses to negative values beyond $N=3$. (b) Sheaf energy contracts geometrically ($E_N/E_0 \approx \rho^N$, $\rho=0.886$) while kernel alignment $\cos(\hat{\mathbf{h}}_N, \ker L_{\mathcal{F}})$ rises monotonically—erasing the low-energy non-harmonic modes that store task semantics.

subspace $\ker L_{\mathcal{F}}$ is left untouched. We make this concrete with two diagnostics. First, the total sheaf energy decays geometrically, $E_N \approx \rho^N E_0$ with an empirical contraction rate $\rho \approx 0.886$. Second—and this is the key point—because the learned coboundary map δ only *imperfectly* separates the perturbation direction from task-discriminative structure, *repeated* contraction eventually removes low-coboundary-energy directions that still carry task signal. As $N \rightarrow \infty$ the state over-aligns with $\ker L_{\mathcal{F}}$ ($\cos(\hat{\mathbf{h}}_N, \ker L_{\mathcal{F}}) \rightarrow 1$), so iterating past a single step erodes the very modes that carry task signal. Table 13 and Figure 8 track activation statistics over 31 target evaluation cells as N varies ($\alpha=1.0$): a single pass removes the bulk of perturbation-induced coboundary noise (recovery peaks at 71.3%) while later steps monotonically raise kernel alignment and collapse recovery to -42.2% at $N=10$. This confirms the strict $N=1$ parameterization.

D GATE BEHAVIOR ON COMPLEX, LONG, AND OUT-OF-DISTRIBUTION CLEAN PROMPTS

We stress-test the divergence gate $\gamma(x) = \sigma((\mathcal{E}_{\mathcal{F}}(x) - \tau)/\mu)$, calibrated on S_{clean} , under two shifts. **(1) Context-length shift.** Appending same-task in-context examples to clean prompts ($T \in \{512, 1024, 2048\}$) across all three models, CS-HLA preserved clean performance in 24 of 27 settings (mean $p_{\text{clean}} = -0.14\%$). The lone edge case (Qwen / task1284 with an RFF kernel) arises because un-normalized RFF energy grows with sequence length ($E_{\text{med}}: 1.7 \rightarrow 44.1$), causing over-correction ($p_{\text{clean}} = -11.26\%$); linear restriction maps show no length-dependent growth, so defaulting long-context deployments to linear maps or length-normalizing $\mathcal{E}_{\mathcal{F}}$ (z-scoring by token count) removes the effect. **(2) Task-distribution shift.** Evaluating deployed gates on clean prompts from 7 unseen tasks (NLI, RTE, ANLI, Financial QA, News Classification, Emotion Detection, Reading Comprehension), Table 14 shows that although OOD content pushes $\mathcal{E}_{\mathcal{F}}$ off the calibration manifold and opens the gate frequently, *a gate opening does not imply degradation*: because the har-

monic update projects along a minimal-norm coboundary direction, the actual clean prediction flip rate stays low (mean 11.6% across models; <3.4% on Qwen) and remains below the un-gated descent flip rate (11.6% vs. 12.9%). The gate is thus a conservative safeguard even under distribution shift.